

Muhammad Usman

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EDUCATION

National University of Computing and Emerging Sciences, Islamabad.

Aug 2022 - June 2026

BSc., Software Engineering [CGPA 3.14]

★ Dean's List Award Spring 2023, Spring 2026

★ [Research Publication](#) | Enhanced RSLP for High-Dimensional Forecasting

EXPERIENCE

RubrixCode

Full Stack Developer

Apr 2025 - Apr 2026

- Contributed to the development of client-facing web applications using [Node.js](#), React, and MongoDB.
- Built backend services and API endpoints supporting core application functionality.
- Designed and optimized database schemas across MongoDB and MySQL.

AIESEC | World's Largest Youth Run Organization

Backend Developer - Network Development Team

Oct 2024 - Feb 2025

- Successfully developed optimized backend features for www.aiesecinpakistan.org/ using Next.js to improve performance and scalability.
- Delivered a **bug-free production deployment** of the platform, which now serves **5,000+ users per semester** for national recruitment applications.
- Contributed to the **national HR Hub** <https://nexusmvp.com/> to digitalize our HR processes.

National University of Computing and Emerging Sciences

Teaching Assistant - Object-Oriented Programming

Sep 2023 - Jan 2024

- Guided **60 students** through OOP concepts, graded assignments, and provided support to the course instructor.

PROJECTS

Final Year Project | Automated App Testing

Aug 2025 - May 2026

- Executed data cleaning and preprocessing scripts to prepare **100,000+** raw customer reviews, removing irrelevant information and formatting text for improved model performance.
- Fine-tuned a **BERT-based** classifier to label reviews as bug reports, feature requests, or usability issues and integrated an inference pipeline to filter actionable feedback.
- Built an **Appium-based automation agent** to dynamically detect app flows, generate actionable test cases from classified feedback, and automatically execute and validate outcomes end-to-end.

FinTech Forecasting | Web Application [Github Repo](#)

Nov 2025 - Dec 2025

- Built a full-stack ML stock prediction platform using FastAPI and Next.js 14 to generate multi-step price forecasts with confidence bands.
- Developed a **15-endpoint** REST API with MongoDB Atlas persistence, real-time portfolio P&L tracking, price alert watchlists, and 5 technical indicator overlays (RSI, MACD, Bollinger Bands, SMA, EMA).

Real-time Collaborative Kanban | Web Application [GitHub Repo](#)

March 2026

- Built a **real-time collaborative** Kanban board using the MERN stack and Socket.io, implementing optimistic UI updates with automatic rollback across 9 bidirectional socket events for instant drag-and-drop responsiveness across concurrent sessions.
- Architected a zero-authentication multi-user board system with share-link access control, cross-list drag-and-drop via dnd-kit, and live activity feed synced via Socket.io rooms.

SKILLS

Programming Languages: C, C++, Python, JavaScript

Web Technologies: React.js, Node.js, Express.js, MongoDB, Next.js, MySQL, Flask, Django, Fast API, REST API